

Solve the following LPP using Simplex

$$Z_{\max} = 3x + 5y$$

$$\text{subject to, } 3x + 2y \leq 18$$

$$0 \leq x \leq 4$$

$$0 \leq y \leq 6$$

Step 1: Find Standard LPP

* Note: For each ($\leq$) type constraint we add (slack) variable to get ($=$) constraints. and coefficient of slack variable is zero in objective function

$$Z_{\max} = 3x + 5y$$

$$C_1: 3x + 2y + S_1 = 18$$

$$C_2: x + S_2 = 4$$

$$C_3: y + S_3 = 6$$

$$\therefore Z_{\max} = 3x + 5y + 0S_1 + 0S_2 + 0S_3$$



x, y (Decision Variable (DV)) ≥ 0 ; S_1, S_2, S_3 (slack) ≥ 0

Step 2: To find initial Basic Feasible solution (IBFS)

$$m \rightarrow (\text{Number of Constraints}) = 3$$

$$n \rightarrow (\text{Number of variables}) = 5$$

To find Basis, we set $(n-m)$ variables equal to zero

$$\therefore (n-m) = 5 - 3 = 2 \text{ variables will take zero values}$$

$$\therefore \text{Set } x = 0, y = 0$$

$$\therefore \text{We get, } S_1 = 18; S_2 = 4; S_3 = 6$$

$$\therefore \text{Basis (Basic variables are) } B = (S_1, S_2, S_3)$$

$$\text{solution of basis } X_B = (18, 4, 6)$$

$$\text{coefficient of basic variables from obj function } C_B = (0, 0, 0)$$

Step 3: Initial simplex table

Coefficients of variables from objective (Z)			RHS					
C_B	B	X_B	x_1	x_2	s_1	s_2	s_3	EQ (rhs)
0	s_1	18	3	2	1	0	0	9
0	s_2	4	0	0	0	1	0	-4
0	s_3	6	0	1	0	0	1	6
$\sum C_B X_B = Z_j$			0	0	0	0	0	
$C_j - Z_j$			3	5	0	0	0	
$Z_{max} = \sum C_B X_B$								

Step 4: 1 improved table

All $C_j - Z_j \neq 0$, solution is not optimal.

Find

① key column and key row is $k_C = \max \{C_j - Z_j / C_j - Z_j > 0\}$
 i.e. That is variable corresponding to the key column
 would enter the basis.

② key row = $k_R : \min \left\{ \frac{X_B}{\text{rve value of } k_C} \right\} \rightarrow ER \text{ (Exchange)}$

i.e. That is key row variable will leave the basis

New Basis = (s_1 , s_2 , x_2) $B = \begin{pmatrix} 0 & 0 & 5 \end{pmatrix}$

key ele = 1

 s_1 s_2 x_2

2. $R_2(\text{New}) = R_2(\text{old})$ (value from kc)

3. $R_1(\text{new}) : R_1(\text{old}) - \frac{(2) R_3(\text{old})}{\text{key_ele}}$

$$2. R_2(\text{new}) = R_2(\text{old}) - \frac{(o) \cdot R_3(\text{old})}{\text{key.ele}} = R_2(\text{old})$$

$$R_1(\text{New}) \cdot \begin{cases} x_B & x_1 & x_2 & s_1 & s_2 & s_3 \\ 18 - 2(6) & 3 - 2(0) & 2 - 2(1) & 1 - 2(0) & 0 - 2(0) & 0 - 2(1) \\ & = 6 & = 3 & 0 & = 1 & = -2 \end{cases}$$

C _j		3	5	0	0	0	0	0	0
C _B	B	X _B	x ₁	x ₂	s ₁	s ₂	s ₃	s ₄	s ₅
0	s ₁	6	3*	0	1	0	0	0	0
0	s ₂	4	0	1	0	1	0	0	0
5	x ₂	6	0	1	0	0	0	0	0
z _{max} = 30	z _j		3	5	0	0	0	0	0

Step 5: 2nd improvement

New Basis : (x_1, s_2, x_2)

$$B = \begin{pmatrix} 3 & 0 & 5 \end{pmatrix}$$

key ele : 3

key row: R_1

1. $R_i(\text{new}) = R_i(\text{old}) - \text{key ele}$ key row = R_i
 $R_i(\text{old}) = 22$ $R_i(\text{old}) = 22$
key ele 3

$$2. R_2(N_{\text{new}}) = R_2(\text{old}) - \frac{(\text{value of } k_c)}{2} R_1(\text{old})$$

$$3 \cdot R_3(\text{New}) = R_3(\text{old})$$

			C_j	3	5	0	0	0
CB	B	x_B		x_1	x_2	s_1	s_2	s_3
3	x_1	2		1	0	$1/2$	0	$2/3$
0	s_2	2		0	0	$1/3$	1	$2/3$
5	x_2	6		0	1	0	0	1
$Z_{max} = 36$	Z_j			3	5	0	0	3
	$C_j - Z_j$			0	0	-1	0	-3

$\therefore$ all $C_j - Z_j \leq 0$, solution is optimal

$$\begin{cases} Z_{max} = 36 \\ x_1 = 2, x_2 = 6 \end{cases}$$

$$OP = 3x_1 + 5x_2 = 36$$

$$2x_1 + 2x_2 + 2s_1 + 2s_2 + 2s_3 = 12$$

Step 3: To find initial basic feasible solution

(1) no. of constraints = 3

(2) no. of variables = 5

(3) no. of variables - no. of constraints = 5 - 3 = 2 variables will take zero value

$$x_1, x_2 = 0$$

$$\therefore 2s_1 + 2s_2 + 2s_3 = 12 \Rightarrow s_1 = 6, s_2 = 6, s_3 = 0$$

Basic variables are $B = (s_1, s_2, s_3)$

Solution of Basis $x_B = (6, 6, 0)$

Coefficient of basic variables objective function $C_B = (0, 0, 0)$

Step 3: Initial Table

C_j	3	5	0	0	0
x_B	x_1	x_2	s_1	s_2	s_3
0	0	0	0	0	0
Ratio					

Solve following LPP using simplex

$$Z_{\max} = 4x_1 + 10x_2$$

$$\text{Subject to, } 2x_1 + x_2 \leq 50$$

$$2x_1 + 5x_2 \leq 100$$

$$2x_1 + 3x_2 \leq 90$$

$$x_1, x_2 \geq 0$$

Step 1: Finding initial LPP

$$Z_{\max} = 4x_1 + 10x_2$$

$$C_1: 2x_1 + x_2 + S_1 = 50$$

$$C_2: 2x_1 + 5x_2 + S_2 = 100$$

$$C_3: 2x_1 + 3x_2 + S_3 = 90$$

$$Z_{\max} = 4x_1 + 10x_2 + S_1 + S_2 + S_3$$

Step 2: To find initial basic feasible solution

m (No. of constraints) = 3

n (No. of variables) = 5

$(n-m) = 5-3 = 2$ variables will take zero values

$$x_1, x_2 = 0$$

$$\therefore S_1 = 50; S_2 = 100; S_3 = 90$$

Basic variables are $B = (S_1, S_2, S_3)$

Solution of Basis $X_B = (50, 100, 90)$

coefficient of basic variables objective function $C_B = (0, 0, 0)$

Step 3: Initial Table

		C_j	4	10	0	0	0	Ratio
C_B	B	X_B	x_1	x_2	S_1	S_2	S_3	
0	S_1	50	2	1	1	0	0	50 $\left\{ \begin{smallmatrix} K_R \\ C_{min} \end{smallmatrix} \right\} \rightarrow$
0	S_2	100	2	5*	0	1	0	20 $\downarrow$
0	S_3	90	2	3	0	0	1	30
			0	0	0	0	0	
$G_j - Z_j$			4	10 $\uparrow$ $\begin{smallmatrix} K_C \\ C_{max} \end{smallmatrix}$	0	0	0	

step 4: 1st improved table

			C_j	4	10	0	0	0
C_B	B	X_B		x_1	x_2	s_1	s_2	s_3
0	s_1	30		$8/5$	0	1	$-1/5$	0
10	x_2	20		$2/5$	1	0	$1/5$	0
0	s_3	30		$4/5$	0	0	$-3/5$	1
$Z_{max} = 200$	$Z_j = \sum C_B X_j$			4	10	0	0	0
	$C_j - Z_j$			0	0	0	-2	0

$$R_2(\text{New}) = \frac{R_2(\text{Old})}{\text{key elem}} = \frac{R_2(\text{Old})}{5}$$

$$R_1(\text{New}) = R_1(\text{Old}) - \frac{R_2(\text{Old})}{5}$$

$$R_3(\text{New}) = R_3(\text{Old}) - (3) \cdot R_2(\text{Old})$$

all $C_j - Z_j \leq 0$, solution is optimal

$$\begin{cases} Z_{max} = 200 \\ x_1 = 0 ; x_2 = 20 \end{cases}$$

Big M - Method

- * Note =
- 1) For $\leq$ type constraints, add slack variable
 - 2) For $\geq$ type constraints, subtract surplus variable and add artificial variable (AV) to get (=) constraint
 - 3) For (=) constraint add artificial variable
 - 4) If objective function ($Z_{\max}$) is maximization coefficient of artificial variable is $(-M)$ and if objective function ($Z_{\min}$) is minimization the coefficient of artificial variable is $(+M)$.
Coefficient of slack and surplus variable is always 0.
(M is a very high positive value)

Solve the following LPP using Big M method

$$Z_{\min} = 50x_1 + 10x_2$$

sub j to $5x_1 + 2x_2 \leq 20$

$$x_1 \geq 3$$

$$x \leq 5$$

$$x_1, x_2 \geq 0$$

Step 1: Initial LPP solution

$$Z_{\min} = 50x_1 + 10x_2$$

$$C_1 = 5x_1 + 2x_2 + S_1 = 20$$

$$C_2 = x_1 - S_2 + A = 3$$

$$C_3 = x_1 + S_3 = 5$$

$x_1, x_2, (DV) \geq 0, S_1, S_3$ (slack) $\geq 0, S_2$ (surplus) $\geq 0, A$ (AV) ≥ 0

$$Z_{\min} = 50x_1 + 10x_2 + 0S_1 + 0S_2 + 0S_3 + MA$$

Step 2:

$$(m-n) = 6-3 = 3; x_1 = 0, x_2 = 0, S_2 = 6$$

Solution: $s_1 = 20$, $A = 3$, $s_3 = 5$, $B = (s_1, A, s_3)$

(0, M, 0)

(20, 3, 5)

		C_j	50	10	0	0	0	M	Ratio
C_B	B	X_B	x_1	x_2	s_1	s_2	s_3	A	
0	s_1	20	5	2	1	0	0	0	4
M	A	3	1*	0	0	-1	0	1	3 $\xrightarrow{KR}$ (min)
0	s_3	5	0	1	0	0	1	0	
$Z_{min} = 3M$		Z_j	M	0	0	-M	0	M	
			50-M	10	0	+M	0	0	
			-95	0	10	0	1000	0	0

* Note: obj (min) For all $c_j - z_j \geq 0$ solution is optimal

$$R_2(\text{New}) = R_2(\text{old})$$

$$R_1(\text{New}) = R_1(\text{old}) - 5(R_2(\text{old}))$$

$$R_3(\text{New}) = R_3(\text{old})$$

When artificial basis leave

basis also we do not consider

it for next improved table

		C_j	50	10	0	0	0	M
C_B	B	X_B	x_1	x_2	s_1	s_2	s_3	A
0	s_1	20	0	2	1	5	0	-5
50	x_1	3	1	0	0	-1	0	
0	s_3	5	0	1	0	0	1	
$Z_{min} = 150$		Z_j	50	10	0	-50	0	M
			0	0	10	0	50	0

$$Z_{min} = 150$$

$$x_1 = 3, x_2 = 0$$

Solve the following LPP using Big M method

$$Z_{max} = 3x_1 - x_2 \text{ subject to,}$$

$$2x_1 + x_2 \geq 2$$

$$x_1 + 3x_2 \leq 3$$

$$x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Step 1: standard LPP

$$Z_{max} = 3x_1 - x_2$$

$$2x_1 + x_2 - S_1 + A_1 = 2$$

$$x_1 + 3x_2 + S_2 = 3$$

$$x_2 + S_3 = 4$$

$$x_1, x_2 \text{ (D.V.)} \geq 0, S_1 \text{ (surplus)} \geq 0; S_2, S_3 \text{ (slack)} \geq 0; A_1 \text{ (A.V.)}$$

$$(n-m) = 6-3 = 3 \text{ (variables 3 will take 0 = surplus)}$$

$$x_1, x_2 = 0, S_1 = 0$$

$$Z_{max} = A_1 = 2, S_2 = 3, S_3 = 4$$

$$B = (A_1, S_2, S_3)$$

$$R = (M=0, 0)$$

$$C = (2, 3, 4)$$

$$Z_{max} =$$

$$C_B \quad B \quad X_B \quad x_1 \quad x_2 \quad S_1 \quad S_2 \quad S_3 \quad A$$

$$-M \quad A_1 \quad 2 \quad 2^*$$

$$0 \quad S_2 \quad 3$$

$$0 \quad S_3 \quad 4$$

$$Z_{max} = -2M$$

$$Z_j$$

$$C_j - Z_j$$

$$3+2M \quad -1+M \quad -M \quad 0 \quad 0 \quad 0 \quad 0$$

$$\uparrow K=999$$

$$200 \quad 3 \quad \text{max} \quad -1000$$

Step : 1st improved table

$$R_1(\text{New}) \Rightarrow \frac{R_1(\text{old})}{\# \text{ key ele}}, \quad \frac{R_1(\text{old})}{2}$$

$$R_3(\text{New}) = R_3(\text{old})$$

$$R_2(\text{New}) = R_2(\text{old}) - \frac{R_1(\text{old})}{2}$$

$$CB = (3, 0, 0)$$

			C_j	3	-1	0	0	0	
CB	B	X_B	x_1	x_2	s_1	s_2	s_3	Ratio	
3	x_1	1	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	1 (-2)	-
0	s_2	2	0	$\frac{5}{2}$	$\frac{1}{2}$ *	1	0	4	
0	s_3	4	0	1	0	0	1	-	
$Z_{\max}:$			3	$\frac{3}{2}$	$-\frac{3}{2}$	0	0		
			0	$-\frac{5}{2}$	$\frac{3}{2}$ $\uparrow$ KCC $Z_{\max}$	0	0		

Step : 2nd Improved table

			C_j	3	-1	0	0	0	
CB	B	X_B	x_1	x_2	s_1	s_2	s_3		
3	x_1	3	1	3	0	1	0		
0	s_1	4	0	5	1	2	0		
0	s_3	4	0	1	0	0	1		
$Z_{\max} = 9$			Z_j	3	9	0	3	0	
			$C_j - Z_j$	0	-10	0	-3	0	

$\therefore$ It's optimal where $C_j - Z_j \leq 0$

$$Z_{\max} = 9$$

$$x_1 = 3 \quad x_2 = 0$$

→ Primal and Dual

Primal

$$1. \quad Z_{\max} = \sum_{j=1}^n C_j x_j$$

subject to m

$$\sum_{j=1}^n a_{ij} x_j \leq b_i \quad (i=1, \dots, m)$$

$$x_j \geq 0 \quad (j=1, \dots, n)$$

x_j primal variables

Dual

$$Z_{\min} = \sum_{i=1}^m b_i y_i$$

subject to n

$$\sum_{i=1}^m a_{ij} y_i \geq C_j$$

$$y_i \geq 0 \quad (i=1, \dots, m)$$

y_i dual variables

2. If we have
- $m \rightarrow$ primal constraints $\rightarrow m \rightarrow$ dual variables
- $n \rightarrow$ primal variables $\rightarrow n \rightarrow$ dual constraints

3. If k^{th} constraint is $(=)$ Then k^{th} dual variable is unrestricted in sign (i.e. -ve, +ve, value)

4. If l^{th} primal variable is unrestricted in sign Then l^{th} dual constraint is $(=)$

* Note: For att $Z_{\max}$ all constraints should be $(\leq)$
 $Z_{\min}$ all constraints should be $(\geq)$
 and all primal and dual variables should be ≥ 0

* Note: $a = b \Rightarrow \begin{cases} a \leq b \\ a \geq b \end{cases}$

Find dual of following primal problems

$$1) Z_{\max} = 3x_1 - x_2 + 7x_3$$

subject to

$$x_1 + 3x_2 - 2x_3 \leq 4$$

$$4x_1 + 2x_2 \geq 6$$

$$x_1, x_2, x_3 \geq 0$$

Step 1: standard primal

$$Z_{\max} = 3x_1 - x_2 + 7x_3$$

subject to

$$x_1 + 3x_2 - 2x_3 \leq 4$$

$$-4x_1 - 2x_2 - 0x_3 \leq -6$$

$$x_1, x_2, x_3 \geq 0$$

$$m \rightarrow 2$$

$$n \rightarrow 3$$

$\rightarrow$ (dual constraints)

(dual variables)

Step 2: Dual

Let y_1, y_2 be required dual variables

$$Z^*_{\min} = 4y_1 - 6y_2$$

$$1y_1 - 4y_2 \geq 3$$

$$3y_1 - 2y_2 \geq -1$$

$$-2y_1 \geq 7$$

$$y_1, y_2 \geq 0$$

$$2) Z_{\min} = -x_1 + 7x_2$$

subject to

$$3x_1 + 2x_2 \geq 5$$

$$x_1 - x_2 \leq 6$$

$$4x_2 \geq 1$$

$$x_1, x_2 \geq 0$$

Step 1: Standard Primal

$$Z_{\min} = -x_1 + 7x_2$$

$$3x_1 + 2x_2 \geq 5$$

$$-x_1 + x_2 \geq -6$$

$$0x_1 + 4x_2 \geq 1$$

$m \rightarrow 3 \rightarrow n \rightarrow 2 \rightarrow$ (dual constraints)
(dual variables)

Step 2: Dual

let y_1, y_2, y_3 be required dual variables.

$$Z^*_{\max} = 5y_1 - 6y_2 + 1y_3$$

$$3y_1 - y_2 + 0y_3 \leq -1$$

$$-y_1 + y_2 + 4y_3 \leq 7$$

$$y_1, y_2, y_3 \geq 0$$

$$3) Z_{\max} = 3x_1 - 4x_2 + 7x_3$$

subject to

$$x_1 + x_2 + x_3 \leq 3$$

$$2x_1 - 6x_2 + 3x_3 = 4$$

$$x_1, x_2, x_3 \geq 0$$

Step 1: Standard Primal

$$Z_{\max} = 3x_1 - 4x_2 + 7x_3$$

$$x_1 + x_2 + x_3 \leq 3$$

$$2x_1 - 6x_2 + 3x_3 \leq 4$$

$$-2x_1 + 6x_2 - 3x_3 \leq -4$$

$$x_1, x_2, x_3 \geq 0$$

$$m = 3 \quad (\text{dual variables})$$

$$n = 3 \quad (\text{dual constraints})$$

Step 2: Dual

Let y_1, y_2, y_3 be required dual variables

$$Z^*_{\min} = 3y_1 + 4y_2 - 4y_3$$

$$y_1 + y_2 - 2y_3 \geq 3$$

$$y_1 - 6y_2 + 6y_3 \geq -4$$

$$y_1 + 3y_2 - 3y_3 \geq 7$$

$$y_1, y_2, y_3 \geq 0$$

As 2nd primal constraint is (=). 2nd dual variable should be unrestricted in sign.

$$\text{Let } y = y_2 - y_3$$

Now as $y_2, y_3 \geq 0 \Rightarrow y$ is unrestricted in sign

$\therefore$ Ans is

$$Z^*_{\min} = 3y_1 + 4y$$

$$y_1 + 2y \geq 3$$

$$y_1 - 6y \geq -4$$

$$y_1 + 3y \geq 7; \quad y_1 \geq 0 \text{ \& } y \text{ unrestricted in sign}$$

$$4) Z_{\min} = x_1 + 3x_2$$

subject to

$$2x_1 + x_2 = 3$$

$$x_2 \leq 4$$

$$x_1 \geq 0 \text{ \& } x_2 \text{ unrestricted in sign}$$

$$\text{Let } x_2 = x_2' - x_2'' \text{ ; where } x_2', x_2'' \geq 0$$

given primal is

$$Z_{\min} = x_1 + 3x_2' - 3x_2''$$

$$2x_1 + x_2' - x_2'' = 3 \rightarrow 2x_1 + x_2' - x_2'' \leq 3$$

$$x_2' - x_2'' \geq 4 \quad \hookrightarrow \quad 2x_1 + x_2' - x_2'' \geq 3$$

$$x_1, x_2', x_2'' \geq 0$$

Standard Primal

$$Z_{\min} = x_1 + 3x_2' - 3x_2''$$

$$2x_1 + x_2' - x_2'' \geq 3$$

$$-2x_1 - x_2' + x_2'' \geq -3$$

$$-x_2' + x_2'' \geq -4$$

$$x_1, x_2', x_2'' \geq 0$$

$$m = 3$$

$$n = 3$$

Dual

Let y_1, y_2, y_3 be required dual variables

$$Z^*_{\max} = 3y_1 - 3y_2 - 4y_3$$

$$2y_1 - 2y_2 \leq 1$$

$$y_1 - y_2 - y_3 \leq 3$$

$$-y_1 + y_2 + y_3 \leq -3$$

$$y_1, y_2, y_3 \geq 0$$

1st primal constraint is (=) $\Rightarrow$ 1st dual variable unrestricted
and 2nd primal variable unrestricted $\Rightarrow$ 2nd dual constraint (=)

Let $y = y_1 - y_2$

Now as $y_1, y_2 \geq 0 \Rightarrow y$ is unrestricted in sign

$$\therefore Z^*_{\max} = 3y - 4y_3$$

$$2y \leq 1$$

$$y - y_3 \leq 3 \quad \left. \begin{array}{l} y - y_3 = 3 \\ -y + y_3 \leq -3 \end{array} \right\}$$

$$-y + y_3 \leq -3$$

y is unrestricted in sign
and $y_3 \geq 0$

Unit 2

Transportation Problems

- * Note: If total supply = total demand then transportation problem is called as balanced otherwise it is called as unbalanced transportation problem.
- > For balanced transportation problem with objective function as Z_{min} following steps are used to find optimal value:

Step 1: To find initial solution

- 1) NWCM (North-West-Corner Method)
- 2) VAM (Vogel's approximation Method)

Step 2: To check the optimality of solution obtained in step 1 we use (u-v) Method or MODI

- > ~~Step 3~~: Non degenerate basic feasible solution:

A basic feasible for $(m \times n)$

Total number of non-negative allocations is exactly $m+n-1$
where $m \rightarrow$ rows $n \rightarrow$ no of columns

Degenerate solutions

no. of allocations $\neq m+n-1$

Example

	D	E	F	Supply
A	2 2	2 2	— 3	4
B	— 2	3 1	3 6	6
C	— 1	— 3	8 4	8
Demand	2	5	11	18

Total cost = 61

$$m+n-1 = 5$$

no. of allocation = 5

Non-degenerative solutions

Step 2: To check the optimality of solution obtained in step 1 we use (u-v) Method or MODI

Step 3: Non-degenerative basic feasible solution:

A basic feasible for (m+n)

Total number of non-negative allocations is exactly $m+n-1$

where m = rows and n = no. of columns

Degenerate solutions

no. of allocations $\neq m+n-1$